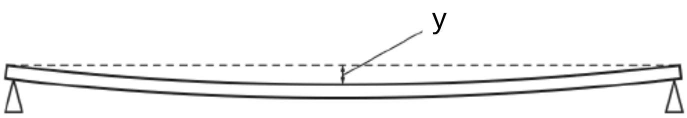
3 A ruler is supported horizontally by two pivots as shown.



The vertical displacement y at the centre of the ruler can be used to measure the mass loaded on it and is given by the equation

$$y = \frac{kML^3}{wt^3}$$

where

k is a constant,
L is the distance between the pivots,
M is the mass loaded onto the ruler,
t is the thickness of the ruler and
w is the width of the ruler.

When a particular M is loaded onto the ruler, the following results are obtained:

$$y = (0.25 \pm 0.01)\ \text{mm}$$
$$L = (80.0 \pm 0.2)\ \text{cm}$$
$$t = (6.0 \pm 0.1)\ \text{mm}$$
$$w = (23.0 \pm 0.5)\ \text{mm}$$

Which measurement contributes the most to the uncertainty of M?

A y **B** L **C** t **D** w

Ans: C

Make M subject of formula:

$$M = \frac{ywt^3}{kL^3}$$

$$\frac{\Delta M}{M} = \frac{\Delta y}{y} + \frac{\Delta w}{w} + 3\frac{\Delta t}{t} + 3\frac{\Delta L}{L}$$

$$\frac{\Delta y}{y} = \frac{0.01}{0.25} = 0.040$$

$$\frac{\Delta w}{w} = \frac{0.5}{23.0} = 0.022$$

$$3\frac{\Delta t}{t} = 3\frac{0.1}{6.0} = 0.050$$

$$3\frac{\Delta L}{L} = 3\frac{0.2}{80.0} = 0.0075$$

4 The speed of an aeroplane in still air is 200 km h^{-1}. The wind pushes it from the west at a speed